

Root Cause Analysis

Asked by : YouTube



Problem Statement : The year is 2020 and you are the product manager at YouTube. You see there is a 20% drop in the Average View Duration. Find the root cause of this ?

Clarifying Questions :

Q1 : What do you mean by Average View duration ?

Ans : The average watch time on a particular video.

Q2 : Which of the YouTube services has been impacted YouTube video sharing, YouTube kids etc. ?

Ans : YouTube video sharing has been impacted.

Q3 : Any specific duration of videos which has been impacted ?

Ans : Video length > 5mins

Q4 : For how long we have seen this drop ?

Ans : Its been a week since we are seeing this drop.

Q5 : Are we seeing this drop across a specific location ?

Ans : No, its worldwide.

Q6 : Are we seeing this drop across a specific demography of people ?

Ans : No, its across all the demographics.

Q7 : Is this drop across app or website ?

Ans : No its across both the platforms.

Q8 : Is this problem industry wide or not ?

Ans : No, its not across other video sharing apps.

Sanity Check :

Q1 : Have we changed the definition of how the metric is calculated ?

Ans : No we haven't changed any definition.

Lets focus on external factors for now, to find out the reason for the impact.

External Factors :

Q1 : Is there any rise of a new competitor ?

Ans : No new competitor has entered the market.

Q2 : Have the existing competitor launched any new features ?

Ans : No

Assumption - I am assuming there are no government regulations on the video sharing apps, as this would have impacted the whole industry.

Q3 : Has there been any good or bad PR against us ?

Ans : No, nothing as such.

Q4 : Have we violated any regulations or promoted something which we shouldn't ?

Ans : No, we have not violated any regulations.

Internal Factors :

Q1 : Have we launched any new feature in the market ?

Ans : Yes we have released a new feature.

Q2 : What is the new feature ?

Ans : This is short videos sections, on which videos which are ≤ 90 secs are present. Creators can create and post these short videos and viewers can watch it. We call this as YouTube Shorts.

Q3 : How is the placement of this new feature on the app ?

Ans : We have created a new section as Shorts.

Q4 : How is the placement of this new feature on the app ?

Ans : We have created a new section as Shorts.

Q5 : What was the aim of this new feature ?

Ans : On analyzing the user behavior, we saw that users attention span is decreasing and we also did an analysis of our competitors like TikTok and felt we should also enter into this market.

Currently we saw that a new feature was launched which could have a significant impact on the watch time. I would also like to explore the customer journey to identify the depth of the impact.

Customer Journey :

User install the app → Opens the app → Lands to the YouTube Feed → User scrolls or searches any video to watch → User watches the video.

Q1 : Is there a decline in the user installing the app ?

Ans : No there is no decline, we even saw an increase in app installation.

Q2 : Is there a decline in DAU or MAU ?

Ans : No there is no decline.

Q3 : Is there a decline in user selecting a video from the YouTube video ?

Ans : Yes there is a decline of 10%.

Q4 : Is there a increase in user dropping off from video after seeing an ad ?

Ans : No

Q5 : Are users exiting long videos earlier than before?

Ans : Yes

Q6 : What are the ways we can watch YouTube Shorts ?

Ans : User can watch through the shorts sections on home page or after searching any particular on search bar, YouTube also recommends shorts related to that particular search.

Q7 : Is there a decline in user searching a video and watching it ?

Ans : Yes there is a decline of 5%.

Q8 : Is there a decline in creators creating long videos ?

Ans : No there has been no decline in that.

Q9 : What % of people are moving to Shorts section after landing on the YouTube main feed ?

Ans : Somewhere around 4-5%

Hypothesis :- The launch of Shorts likely changed user intent at session start.

Instead of:

- "Watch one long video"
Users now: Watch Shorts
- Reduce patience for long intros
- Exit long videos earlier

This explains:

- Earlier drop-offs
- Lower Average View Duration even when videos start

The 20% drop in Average View Duration for long videos is primarily driven by a shift in user consumption behavior post the launch of Shorts.

While Shorts adoption is only ~4–5%, it changes session intent and attention patterns, leading to:

- Reduced selection of long videos
- Earlier exits from long-form content
- Lower tolerance for slow pacing

Shorts is not cannibalizing traffic directly, but **fragmenting attention**.

Based on the above user journey analysis, its clear that YouTube Shorts is the reason for drop in Average View duration of videos > 5 mins.